

Summary of EMT Tools for IBR Studies

This document summarizes some of the capabilities and learning paths for new users of software tools for electromagnetic transient (EMT) studies of inverter-based resources (IBRs). NERC does not endorse any of these products, and the list of tools is not exclusive. The tool developers provided this information, which is presented in alphabetical order. All links are publicly accessible.

Table of Contents

EMTP	1
HYPERSIM	2
Kestrel EMT	3
ParaEMT	4
PSCAD	5
RTDS	6

EMTP

EMTP® is a simulation software which includes 3 solvers:

- multiphase unbalanced load-flow
- phasor-domain steady-state
- iterative time-domain (EMT) relying on Trapezoidal and Backward Euler integrations, initialized from load-flow results.

Because EMTP has both load-flow and EMT solvers, it is very easy for new users to create and simulate new design, or design imported from third party formats like PSS/E or CIM. There is no need to have expertise in how to initialize the simulations and the combination of Trapezoidal and Backward Euler integrations and the iterative solver prevent the majority of possible numerical issues allowing user without expertise to perform advance simulations.

EMTP has a wide range of models for inverter-based resources (IBR) in its libraries, including photovoltaic systems (cells, panels, and entire parks), wind power generation (individual turbines and parks of type 3 and type 4), battery energy storage systems, FACTS, HVDC, and more. These systems can be represented as detailed models, incorporating individual wind turbines or PV modules, converters, step-up transformers, collector grids, and control and protection systems, or as aggregated models, depending on the study requirements. The built-in IBR models in EMTP are customizable and automatically adapt to load-flow, time-domain, and frequency-scan simulations without user intervention. All parameters are documented with tooltips directly in the user interface and comprehensive documentation is also available in PDF format.

EMTP also has built-in tools to facilitate and automate grid code compliances and verifications of IBR models, to manage network model versioning and contingencies.

100 online videos and tutorials are also available to learn the first steps, more advanced studies or to see user cases. A dedicated online school is also available with recorded courses and questions between courses.

The EMTP support team also assists every new user in their first step and answers any questions along their journey.

EMTP resources:

- Overview for new users: <https://www.emtp.com/documentation/emtp-overview>
- User manual: <https://www.emtp.com/documentation/emtp-user-manual>
- Online university: <https://emtp.learnworlds.com/>
- Tutorials and user presentations: <https://www.youtube.com/@EMTP-software>
- WECC Wind IBR Model: <https://emtp.com/documents/EMTP-Documentation/doc/renewables/wecc-wind-park.pdf>
- WECC Solar IBR Model: <https://www.emtp.com/documents/EMTP-Documentation/doc/renewables/wecc-pv-park-models.pdf>
- HVDC-MMC Model: <https://www.emtp.com/documents/EMTP-Documentation/doc/hvdc/mmc-documentation-v2.pdf>

HYPERSIM

HYPERSIM¹, commercialized by OPAL-RT, is an EMT simulation tool that utilizes a modified nodal analysis approach for the solution of the network equations and leverages parallel processing to automatically distribute the computational load of large-scale power system models onto various cores of modern multi-core CPUs to achieve real-time simulation performance. This is done by automatically analyzing the network model to identify long transmission lines where the propagation delays are greater than the simulation time-steps to decouple the system model into smaller chunks to achieve real-time performance. HYPERSIM also utilizes an interpolation technique to accommodate accurate simulation of switching elements and an iterative solver for non-linearities such as transformers, surge arresters, etc. HYPERSIM can be downloaded online² and detailed documentation about all things from installation to component library and examples are available in the wiki³. HYPERSIM also has an online learning platform with a self-paced course that they can sign-up for free to familiarize them with the various capabilities and features of the tool⁴. In the context of IBR integration studies and from a new user's perspective, typically there are

¹ HYPERSIM Capabilities brochure: https://blob.opal-rt.com/medias/L00161_0245.pdf

² HYPERSIM Download link: <https://www.opal-rt.com/download-center/#hypersimlink>

³ HYPERSIM Wiki: <https://opal-rt.atlassian.net/wiki/spaces/PDOCHS/overview>

⁴ Learning portal for HYPERSIM: <https://www.opal-rt.com/online-learning/>

a few key steps as outlined in the presentation⁵ to leverage HYPERSIM for performing large-scale EMT studies in real-time.

- For the model migration/conversion from existing phasor domain tools or other EMT tools, HYPERSIM has several examples that demonstrate how this can be performed^{6,7}.
- HYPERSIM has various approaches (user coded model-based⁸, functional mockup unit-based, IEEE/CIGRE B4.82 guideline compatible black-box⁹, etc.,) to integrate detailed, site-specific, OEM black-box IBR plant models with a large network model efficiently.
- HYPERSIM also comes with an example that uses its Python API to show new users how the API can be used to automate the compilation, execution of repeated simulations with varying parameters, and for recording/analyzing simulation datasets¹⁰. This Python API can also be leveraged to automate model quality validation tests for IBR interconnection studies.

When building models for detailed studies, users can also model the switching stages of power electronics converters or HVDC converters such as Modular Multilevel Converters (MMCs) inside the FPGA and co-simulate with their corresponding controls and network models in the CPU¹¹. Additionally, HYPERSIM can also be used to perform real-time, hybrid EMT-Phasor simulations, where a portion of the large network can be simulated in EMT with the remainder simulated in the phasor domain by utilizing OPAL-RT's ePHASORSIM solver as a functional mockup unit inside the HYPERSIM model¹².

Kestrel EMT

Kestrel EMT is a freely available, general purpose EMT simulator¹³ that is focused on making illustrative EMT studies accessible and easy to set up for new users and students. The software package is downloadable from the web site¹⁴ and includes the EMT simulator itself, a circuit designer GUI, a plotting tool (KPlot), and a handful of example study cases. As of January 2025,

⁵ HYPERSIM Workflow: https://emtworkshop.ornl.gov/wp-content/uploads/2023/09/3.-Opal_RT_EMT_Tools_EMT_Workshop.pdf

⁶ Example showing PSSE import to HYPERSIM: <https://opal-rt.atlassian.net/wiki/spaces/PDOCHS/pages/416711042/Examples+PSS+E+Import+IEEE+118-Bus+System>

⁷ Example showing PSCAD import to HYPERSIM: <https://opal-rt.atlassian.net/wiki/spaces/PDOCHS/pages/286032566/Examples+PSCAD+Import+IEEE+39-Bus+System>

⁸ HYPERSIM User Coded Model to include black-box models: <https://opal-rt.atlassian.net/wiki/spaces/PDOCHS/pages/150373966/UCM+Protecting+Code+and+Intellectual+Property+IP>

⁹ HYPERSIM CIGRE interface to include black-box models: <https://opal-rt.atlassian.net/wiki/spaces/PDOCHS/pages/1150812538/CIGRE+Interface>

¹⁰ Example showing HYPERSIM's Python API: <https://opal-rt.atlassian.net/wiki/spaces/PDOCHS/pages/150078785/Python+API+Examples>

¹¹ Example showing FPGA/CPU co-simulation using half-lines: <https://opal-rt.atlassian.net/wiki/spaces/PDOCHS/pages/886898944/Example+FPGA-Based+Multirate+stubline+for+HYPERSIM+eHS>

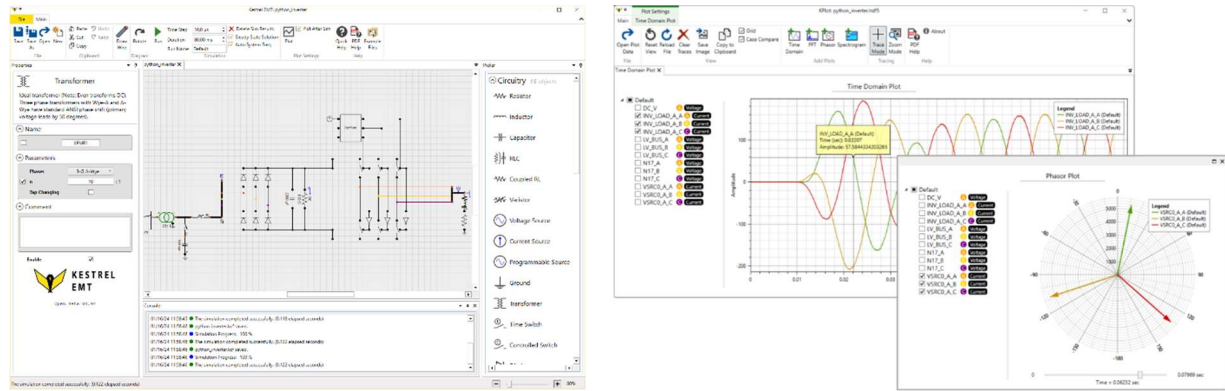
¹² HYPERSIM EMT-Phasor Simulation: https://emtworkshop.ornl.gov/wp-content/uploads/2024/08/1.-Opal_RT_Zerui_Dong.pdf

¹³ Dommel, H.W. (1986) EMTP Theory Book. Bonneville Power Administration, Portland.

¹⁴ Kestrel EMT Website: <https://kestrel-solver.com/>

Kestrel EMT has approximately 40 built in models, including basic RLC components, transmission lines, transformers, and digital logic blocks. Python and MATLAB blocks can also be included in the simulation to rapidly prototype and interface external IBR control code with the EMT simulation. Implementation of WECC inverter control models, a frequency dependent line model, and additional example study cases is slated for the next release in mid-2025. The public version of Kestrel EMT supports Microsoft Windows, and a Linux version is available upon request.

New user documentation is bundled with the installer or available separately¹⁵.



ParaEMT

Time performance: ParaEMT is an open-source electromagnetic transients (EMT) simulator developed at NREL to simulate and analyze transient behaviors of large-scale power systems with inverter-based resources (IBRs)^{16,17,18}. Its Python-based implementation, combined with just-in-time compilation using Numba, enables it to complete a 1-second EMT simulation with a 50-microsecond time step for the WECC 240-bus system in 29 seconds on a single CPU¹⁷. By leveraging parallel computing on high-performance clusters, ParaEMT provides a scalable solution, achieving up to tenfold speedup for systems with thousands of buses^{16,17,19}.

EMT and IBR modeling: ParaEMT employs the conventional nodal formulation-based EMT simulation strategy²⁰ and integrates a generic IBR model^{21,22} to emulate IBR-related dynamics. It

¹⁵ Kestrel EMT and KPlot Documentation: https://kestrel-solver.com/docs/Kestrel_EMT_Documentation_Latest.pdf

¹⁶ M. Xiong, B. Wang, D. Vaidhyanathan, J. Maack, M. Reynolds, A. Hoke, K. Sun, J. Tan, "ParaEMT: an open source, parallelizable, and HPC-compatible EMT simulator for large-scale IBR-rich power grids," *IEEE Trans. Power Del.*, vol. 39, no. 2, pp. 911-921, Apr. 2024.

¹⁷ M. Xiong, B. Wang, D. Vaidhyanathan, J. Maack, M. Reynolds, A. Hoke, K. Sun, D. Ramasubramanian, V. Verma, J. Tan, "An open-source parallel EMT simulation framework," *Electric Power Syst. Res.*, vol. 235, 2024, Art. no. 110734.

¹⁸ ParaEMT GitHub Repository: http://github.com/NREL/ParaEMT_public

¹⁹ S. Fan, H. Ding, A. Kariyamasam, and A. M. Gole, "Parallel electromagnetic transients simulation with shared memory architecture computers," *IEEE Trans. Power Del.*, vol. 33, no. 1, pp. 239-247, Feb. 2018.

²⁰ H. W. Dommel, *EMTP Theory Book*. Portland, OR, USA: Bonneville Power Admin., 1986.

²¹ P. Pourbeik, "Model user guide for generic renewable energy system models," Elect. Power Res. Inst., Palo Alto, CA, USA, Tech. Rep. EPRI 3002006525, 2015.

²² EPRI, "Code Based Generic Inverter Based Resource Model", Nov. 2023.

also supports co-simulation using the HELICS co-simulation interface²³ and integration with third-party IBR models via dynamic linked library (DLL) interfaces²². Currently, the HELICS and DLL interfaces are case-specific and require further development to become more generalized. ParaEMT initializes EMT simulations from solved power flow cases and provides features such as result down-sampling and simulation snapshots.

Support: To support new users, ParaEMT includes preconfigured test systems, including IEEE benchmark cases and WECC systems¹⁸, and provides gradual learning of the simulation process and underlying code through its detailed documentation (currently under development, ~80% complete).

Contacts: Jin.Tan@nrel.gov, Andy.Hoke@nrel.gov, Min.Xiong@nrel.gov

PSCAD

Model Library and Support for IBR Studies: PSCAD provides a comprehensive library of power electronic inverter models for power systems, allowing users to customize inverter topologies. Most renewable energy inverter manufacturers already have their specific OEM models available in PSCAD. As a result, PSCAD is widely used by utilities to analyze the integration of inverter-based resources (IBRs) into the power grid. Additionally, PSCAD offers an extensive knowledge base²⁴, including example cases, detailed documentation, and recorded video webinars. These resources create a seamless learning experience, enabling users to quickly become proficient in conducting interconnection studies.

Interactive Simulations: A standout feature of PSCAD is its ability to conduct interactive simulations. Users can adjust parameters—such as changing gains or applying faults—using built-in controls like sliders, switches, and dials while observing the simulation results continuously. This interactive capability is invaluable for investigative studies, system design, and educational purposes. It is one of the most unique and valuable features of the PSCAD user interface.

Automation and Parallelization: PSCAD offers a comprehensive Python interface for automating IBR studies using batch scripts. To further streamline automation, the free GUI-based software PSAPPS²⁵ is available. PSAPPS automates tasks such as setting up simulations, running multicore parallel processing of PSCAD simulation sets, collecting results, and generating reports in MS Word. By eliminating the need to learn Python programming, PSAPPS helps accelerate the adoption of PSCAD automation.

PSCAD also supports multi-rate, multi-version OEM models for simulations. Built-in parallel network interface (PNI) allows multiple plants or feeders to run in parallel within a single simulation. To further enhance simulation speed and efficiency, the built-in current-scaling component enables accurate aggregation of IBRs. All these powerful features are included in every PSCAD package.

²³ HELICS: <https://helics.org/>

²⁴ <https://www.pscad.com/knowledge-base>

²⁵ <https://www.nayakcorp.com/psapps>

Enerplot for Plot Benchmarking: PSCAD integrates seamlessly with Enerplot²⁶, a robust tool for visualizing and analyzing simulation results from multiple power system digital simulators. Users can benchmark simulations from PSCAD, RTDS, and PSSE by scripting workflows using the embedded Python interface or recording actions for automation. This integration streamlines renewable energy interconnection benchmarking and ensures consistency across various simulation platforms.

Import from other Simulators: PRSIM is a PSCAD companion tool designed to facilitate the import of cases from the PSSE and Power Factory XML data format. PRSIM can systematically import large cases (with thousands of buses) and provide a graphical representation of the imported circuit. It enables seamless initialization of PSCAD using power flow data and automatically updates PSCAD with new load flow results.

Standardized Testing for Compliance: Several utilities have developed and distributed their standardized templates and testing components in PSCAD. They streamline compliance processes, ensuring IBR models meet their exacting quality standards and improve the compliance workflow. PSCAD's open design is purposefully built for supporting such community-driven effort.

RTDS

The RTDS® Simulator, provided by RTDS Technologies, is a real-time power system simulator capable of EMT and hardware-in-the-loop studies for modelling and testing IBRs, their associated controls, and their impact on the power system. The RTDS Simulator consists of modular, custom-integrated simulation hardware that can be sized appropriately to meet the simulation needs of the user, ranging from small microgrids to large-scale transmission system simulation, and an all-in-one simulation software called RSCAD® FX²⁷. The RTDS Simulator has many features enabling IBR studies, including:

- Modelling libraries containing a wide range of scalable plant models, including solar PV, wind turbine, battery energy storage, and hydrogen fuel cells.
- The ability to model power electronic converters in detail at a wide range of simulation timesteps (~2-50 µs) while tightly connected with the surrounding AC network. External PWM controls can be tested, with frequency components in the ~150 kHz range captured by the simulation²⁸.
- The ability to integrate black-boxed vendor models into the real-time simulation²⁹. Replacing generic IBR control models with vendors' real code in a black-boxed software-in-the-loop setup gives system operators a more accurate picture of IBR dynamic behaviour while protecting the intellectual property of the manufacturer.
- A built-in frequency analysis tool allows for the convenient assessment of grid stability. Users can identify harmonics, resonances, and oscillations in the presence of power

²⁶ <https://www.pscad.com/software/enerplot/overview>

²⁷ RTDS Simulator technology overview: <https://www.rtds.com/technology>

²⁸ Universal Converter Model description: <https://www.rtds.com/blogposts/ucm>

²⁹ Black box control integration with the RTDS Simulator: <https://www.rtds.com/news-and-events/rtds-events/webinars/webinar-black-box-control>

electronic converters and estimate the short circuit capacity, which has a strong influence on the performance of integrated IBRs³⁰.

- Documented sample simulation cases focusing on IBR interconnection testing including IEEE 1547™ compliance, low- and high-voltage ride-through testing (LVRT/HVRT), power plant controller (PPC) testing, grid-forming control testing, and inverter testing via power-hardware-in-the-loop.

RTDS Simulator learning paths include free webinars, in-person training opportunities, a searchable online Knowledge Base³¹, and a library of training videos accessible for users, including software walk-throughs and hardware-in-the-loop demonstration videos.

³⁰ The RTDS Simulator's frequency analysis tool: <https://www.rtds.com/blogposts/frequencyanalysis>

³¹ RTDS Simulator online Knowledge Base: <https://knowledge.rtds.com/>